

Saad Alif A Zaman

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RESEARCH INTERESTS

• Computer Vision for Industrial & Infrastructure Inspection • Deep Learning under Data Scarcity • AI-Based Monitoring for Marine, Environmental & Structural Systems • Applied Perception for Real-World Deployment

RESEARCH EXPERIENCE

➤ **Research Assistant** (August 2025 – Present)

“Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection”

Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept of Naval Architecture and Marine Engineering, BUET
Samiul Based Shuvo, Assistant Professor, Dept of Biomedical Engineering, BUET

- Addressing a core bottleneck in applied computer vision, the lack of large labeled datasets in real-world industrial settings, we proposed a k–n Fold Augmentation Cross-Validation (k–n Fold ACV) protocol, together with a novel Interaction-Effect (IE) Metric that quantifies order-sensitivity among data augmentations. The framework was validated on a newly curated NDBD dataset of 35 high-resolution ship-hull images containing diverse biofouling patterns. Eleven domain-specific augmentations and 110 pairwise combinations were analyzed using YOLOv8m-seg as the base model, with cross-validation on YOLO11m-seg and Mask RCNN confirming generalizability. The [code and dataset](#) are publicly available.

➤ **Undergraduate Thesis** (June 2023 – July 2024)

“Developing a biofouling & corrosion identification and distribution system using video images based on deep learning”

Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept of Naval Architecture and Marine Engineering, BUET

- Collected and curated a biofouling & corrosion dataset from Narayanganj Dockyard, Bangladesh and trained YOLOv8 segmentation variants (transfer learning) to detect and localize biofouling and steel-corrosion defects using video/images (fouling F1 up to 0.90); implemented annotation (CVAT → YOLO), cloud training (Colab/Kaggle), and produced a real-time detection prototype for automated hull inspection, reducing manual labor and enabling sustainable maritime maintenance by minimizing fuel consumption and emissions.

➤ **Junior Year Project** (May 2022 – May 2023)

“Designing an Inland 100 Passenger Ship for Inland waterways of Bangladesh”

Supervisor: Dr. Md. Mashiur Rahaman, Professor, Dept of Naval Architecture and Marine Engineering, BUET

- Designed a conceptual 100-passenger inland vessel with an electric propulsion system tailored to Bangladesh’s inland waterway transport needs to reduce CO₂ emissions and improve energy efficiency compared to conventional diesel systems.

PUBLICATIONS

- Minhazul Islam, **Saad Alif-A Zaman**, Md. Mashiur Rahaman, “[Concept of an Electric-Powered Passenger Vessel in Bangladesh](#),” Journal of Maritime Research, vol. 20, no. 3, pp. 68–71, Dec. 2023

EDUCATION

B.Sc. in Naval Architecture And Marine Engineering ([NAME](#))

(April 2019 – July 2024)

Bangladesh University of Engineering and Technology ([BUET](#))

Cumulative GPA: 3.37 / 4.00

RESEARCH GRANTS, RECOGNITION & SCHOLARSHIPS

- RISE Student Research Grant 2023:**
Awarded a competitive Student Research Grant (BDT 65,790) by Research and Innovation center for Science and Engineering ([RISE](#)) for the research project: “Developing a biofouling & corrosion identification and distribution system using video images based on deep learning.”
- SPARRSO Research Fellowship Program 2025–26:**

Space Research and Remote Sensing Organization ([SPARRSO](#)), Bangladesh, shortlisted for final round of competitive research fellowship and invited to a PowerPoint Presentation session, presenting the proposal: *"Development of a Fixed-Wing VTOL Quad Plane for Flood Monitoring & Disaster Response."*

- **University Stipend Scholarship-2019, 2020, 2021 – BUET:**
Competitive merit-based financial award granted to high-performing undergraduate students based on academic results.
- **Rajshahi Board Scholarship HSC 2018:**
Merit-based national scholarship awarded for outstanding academic performance in the Higher Secondary Certificate (Grade 12 equivalent) examination.

STANDARDIZED TEST SCORES

International English Language Testing System (IELTS) Academic

(15th December, 2025)

Overall Band Score: 7.5 (Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 7.0)

WORK EXPERIENCE & INTERNSHIP

• Materials Development Instructor

Udvash Academic & Admission Care | Dhaka, Bangladesh

(April 2025 – August 2025)

Developed and reviewed assessment materials for SSC, HSC, engineering, and university admission examinations. Evaluated question quality, solution accuracy, and curriculum alignment. Contributed to model test preparation and academic content development.

• Academic Mentor (Mathematics & Physics)

(2019 – Present)

Provided one-to-one and small-batch instruction to secondary and higher-secondary students (Bangla and English medium curricula), from foundational concepts through advanced problem solving. Multiple students subsequently admitted to BUET, KUET, DU, and IUT.

• Industrial Internship – Shipyard Practice

Khulna Shipyard Limited | Khulna, Bangladesh

(October 2022 – November 2022)

TECHNICAL SKILLS

- **Programming:** Python, C, C++
- **Deep Learning models:** Ultralytics YOLO, Mask R-CNN
- **Data Annotation Tools:** CVAT, Roboflow, Labeling
- **Data Processing:** Roboflow (augmentation pipelines, preprocessing)
- **Research Computing & Collaboration:** Google Colab, Kaggle
- **Frameworks:** PyTorch, TensorFlow
- **Document Preparation:** LaTeX, Microsoft Office Packages
- **CAD software:** AutoCAD, Rhino
- **Finite Element Analysis software:** Abaqus

LEADERSHIP & VOLUNTEERING

• Founder, Monthly Reading Initiative (Telegram):

(March 2024 – Present)

Founded a monthly book and magazine reading platform on Telegram, which continues to be maintained by junior students and now includes regular quizzes and discussion activities.

• Lead Organizer, Hall Cultural Decoration Initiative:

(March 2024, Kazi Nazrul Islam Hall, BUET)

Led a student initiative to decorate the residential hall, coordinating lighting and visual displays to foster cultural engagement and community spirit in the final year of undergraduate study.

• Badhon BUET Zone, Volunteer Member:

(May 2019 – July, 2024, Dhaka)

Active volunteer in a student-led blood donation organization; donated blood multiple times and participated in donation drives and awareness activities.

REFERENCES

Dr. Kazi Naimul Hoque

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